

## Calibration results

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### Normalized Residuals

Reprojection error (cam0): mean 0.632014795302699, median 0.48049470547305106, std: 0.5451479444859509  
Gyroscope error (imu0): mean 1.1569144573387615, median 1.016090745503132, std: 0.6654450321294149  
Accelerometer error (imu0): mean 1.0506745164276539, median 0.9444752855665973, std: 0.5980591471890411

### Residuals

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Reprojection error (cam0) [px]: mean 0.632014795302699, median 0.48049470547305106, std: 0.5451479444859509  
Gyroscope error (imu0) [rad/s]: mean 0.005809977656586451, median 0.005102766666100054, std: 0.0033418380622987297  
Accelerometer error (imu0) [m/s<sup>2</sup>]: mean 0.02403781326240413, median 0.021608138572350107, std: 0.013682671346100671

### Transformation (cam0):

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T\_ci: (imu0 to cam0):

```
[[-0.99934057  0.02320852  0.02792469  0.05096412]
 [ 0.02794872  0.0007119  0.9996091  0.04900398]
 [ 0.02317957  0.99973039 -0.00136008 -0.00490746]
 [ 0.          0.          0.          1.          ]]
```

T\_ic: (cam0 to imu0):

```
[[ -0.99934057  0.02794872  0.02317957  0.04967467]
 [ 0.02320852  0.0007119  0.99973039  0.00368844]
 [ 0.02792469  0.9996091 -0.00136008 -0.05041466]
 [ 0.          0.          0.          1.          ]]
```

timeshift cam0 to imu0: [s] ( $t_{imu} = t_{cam} + \text{shift}$ )

0.014895025043715156

Gravity vector in target coords: [m/s<sup>2</sup>]

[ 0.10572829 -9.72169104 -1.28295276]

## Calibration configuration

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cam0

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Camera model: pinhole

Focal length: [601.1733383650345, 600.2663647399592]

Principal point: [314.0164516341506, 250.039444021504]

Distortion model: radtan

Distortion coefficients: [0.139268662386042, -0.30741279731381366, 0.0007272799549728076, 0.0006655761403425819]

Type: aprilgrid

Tags:

Rows: 5

Cols: 10

Size: 0.035 [m]

Spacing 0.0105 [m]

## IMU configuration

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IMU0:

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Model: calibrated

Update rate: 125

Accelerometer:

Noise density: 0.002046311532059803

Noise density (discrete): 0.0228784584441373

Random walk: 3.2793722442857696e-05

Gyroscope:

Noise density: 0.0004491777211520533

Noise density (discrete): 0.005021959592372182

Random walk: 1.4932100720276404e-05

T\_ib (imu0 to imu0)

[[1. 0. 0. 0.]

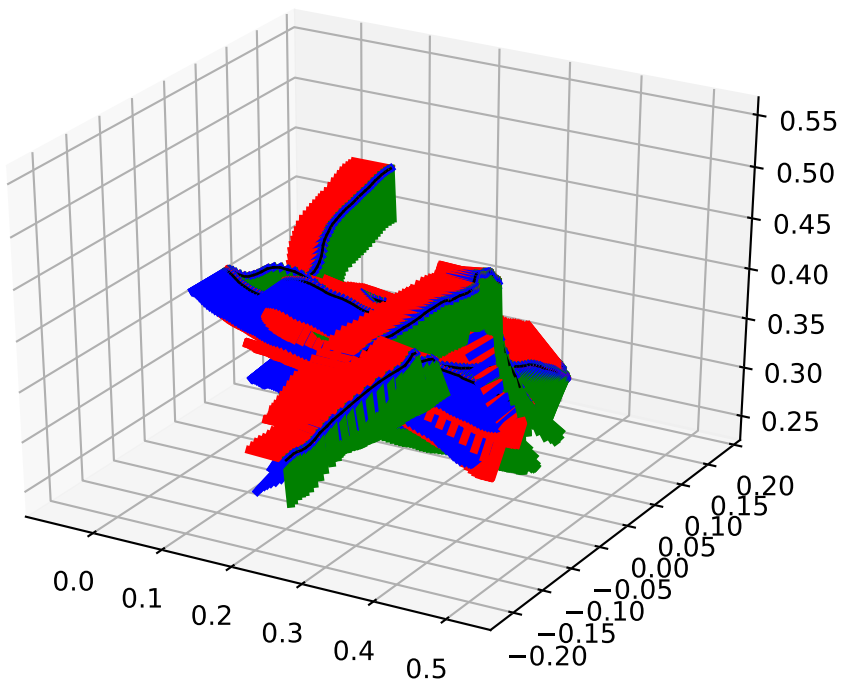
[0. 1. 0. 0.]

[0. 0. 1. 0.]

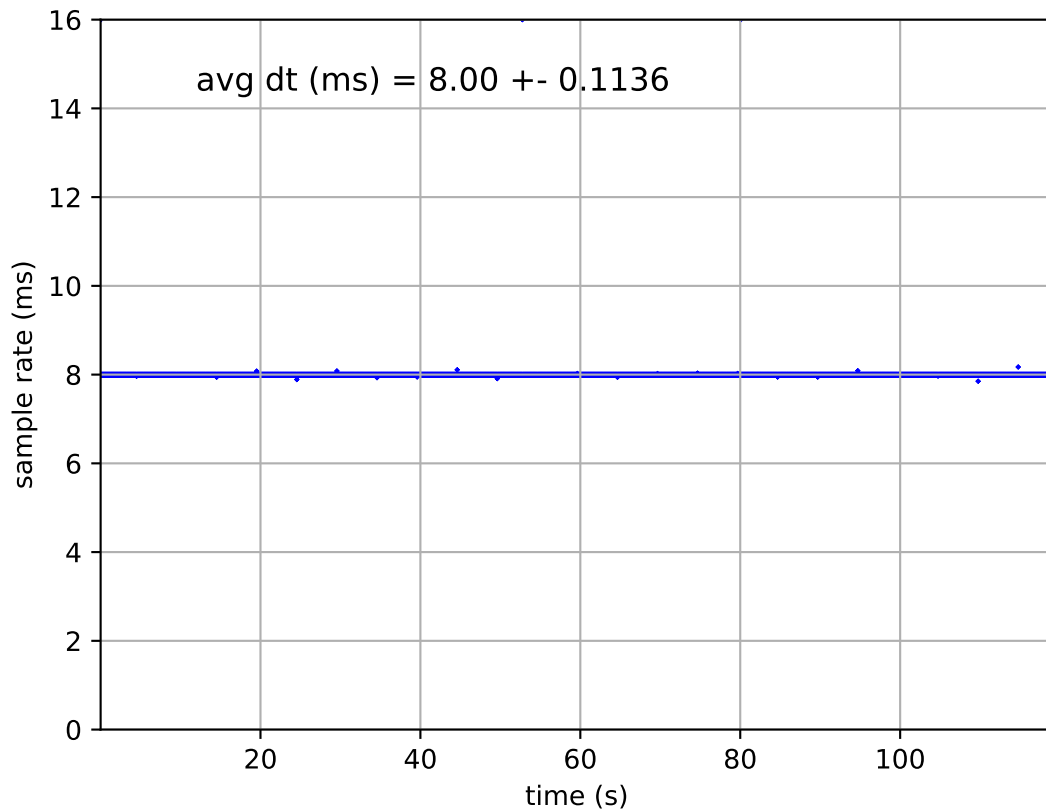
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

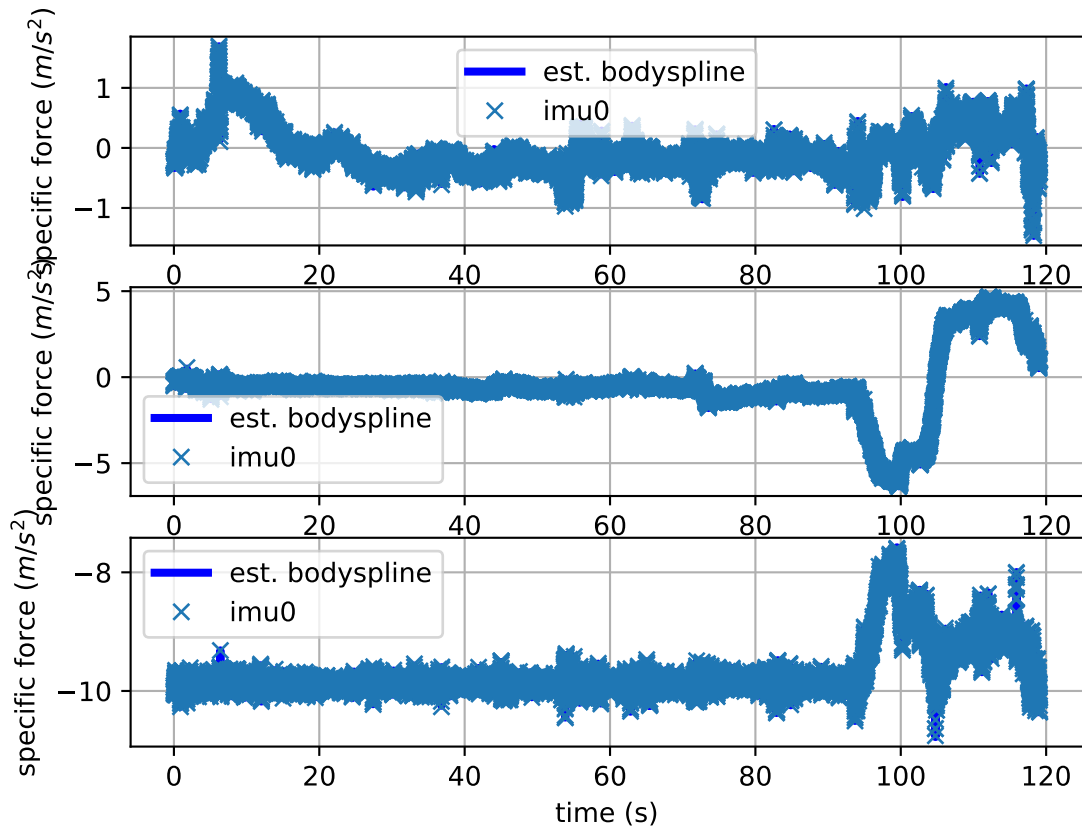
imu0: estimated poses



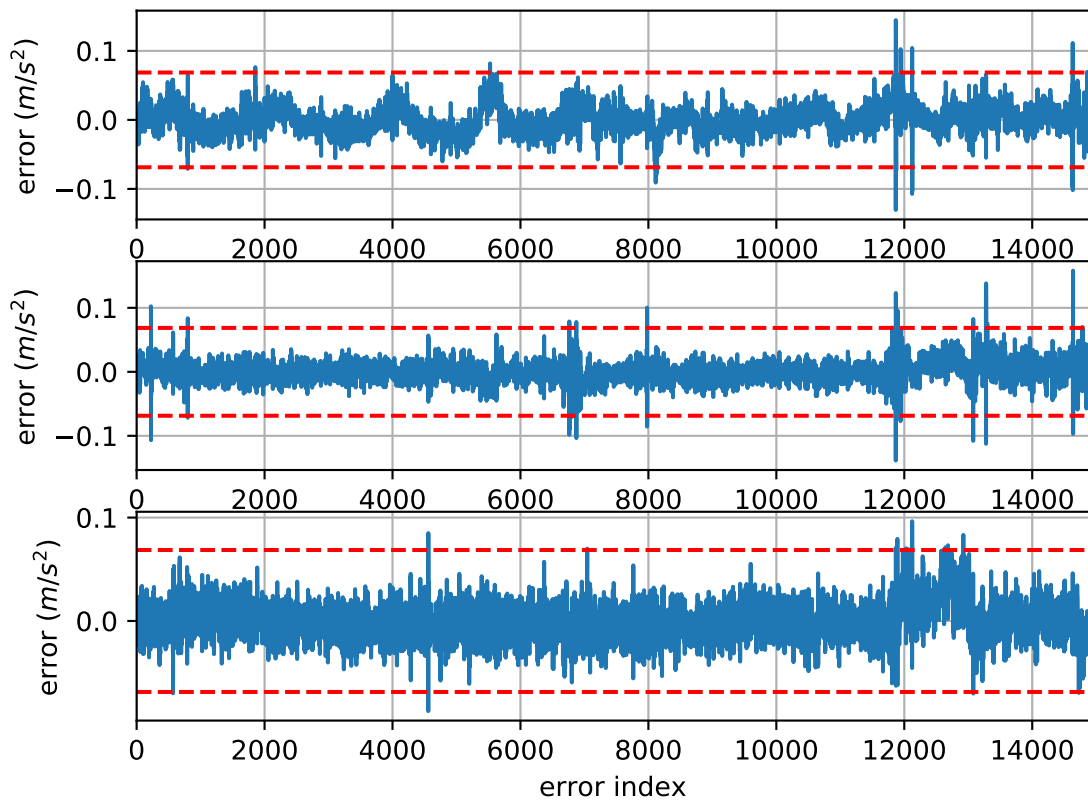
imu0: sample inertial rate



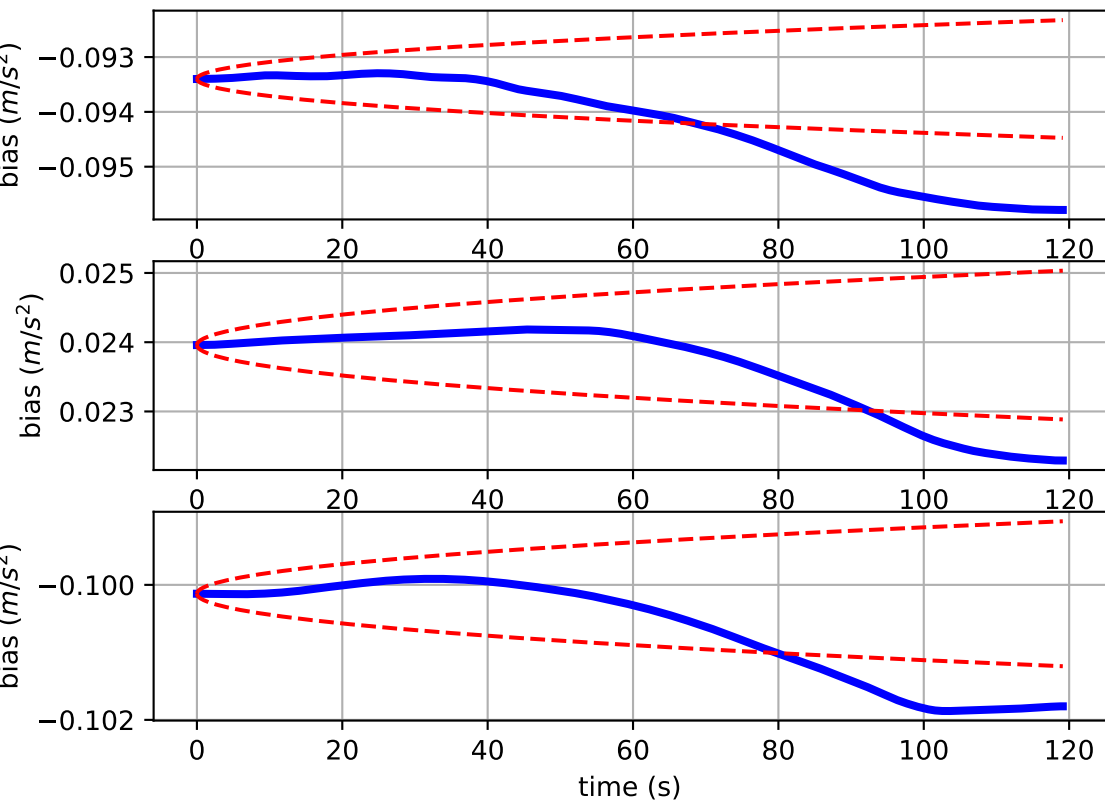
Comparison of predicted and measured specific force (imu0 frame)



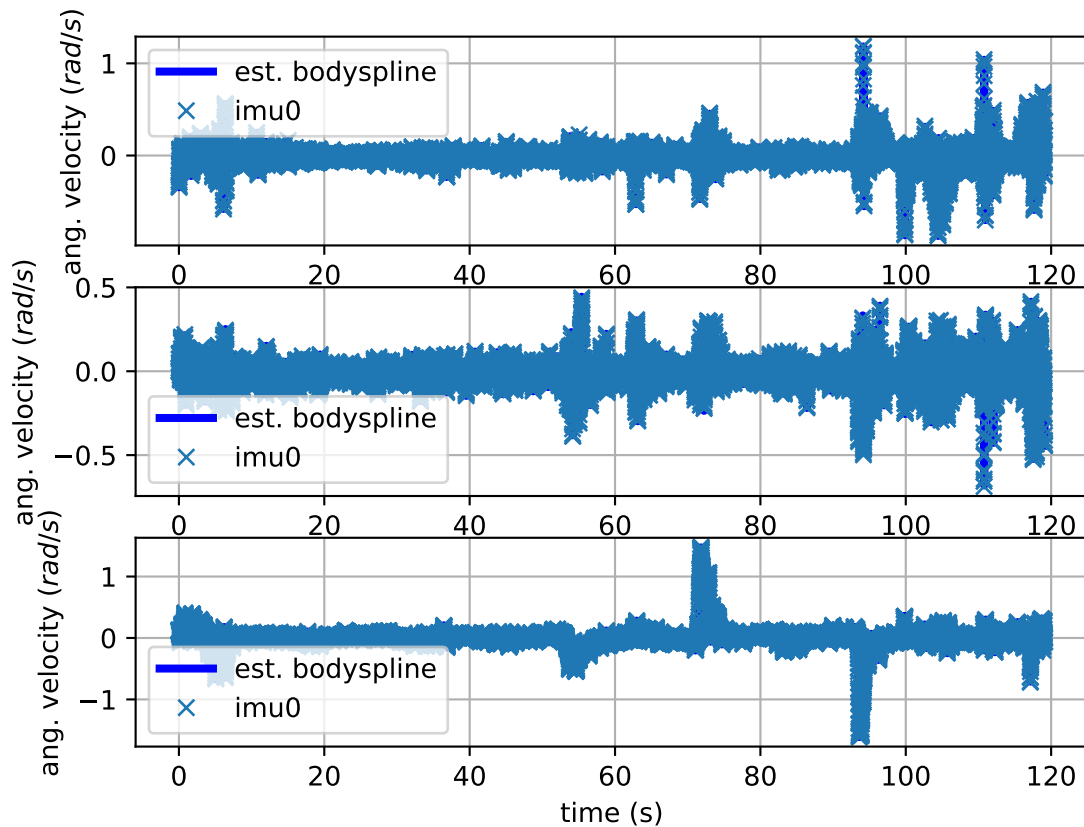
imu0: acceleration error



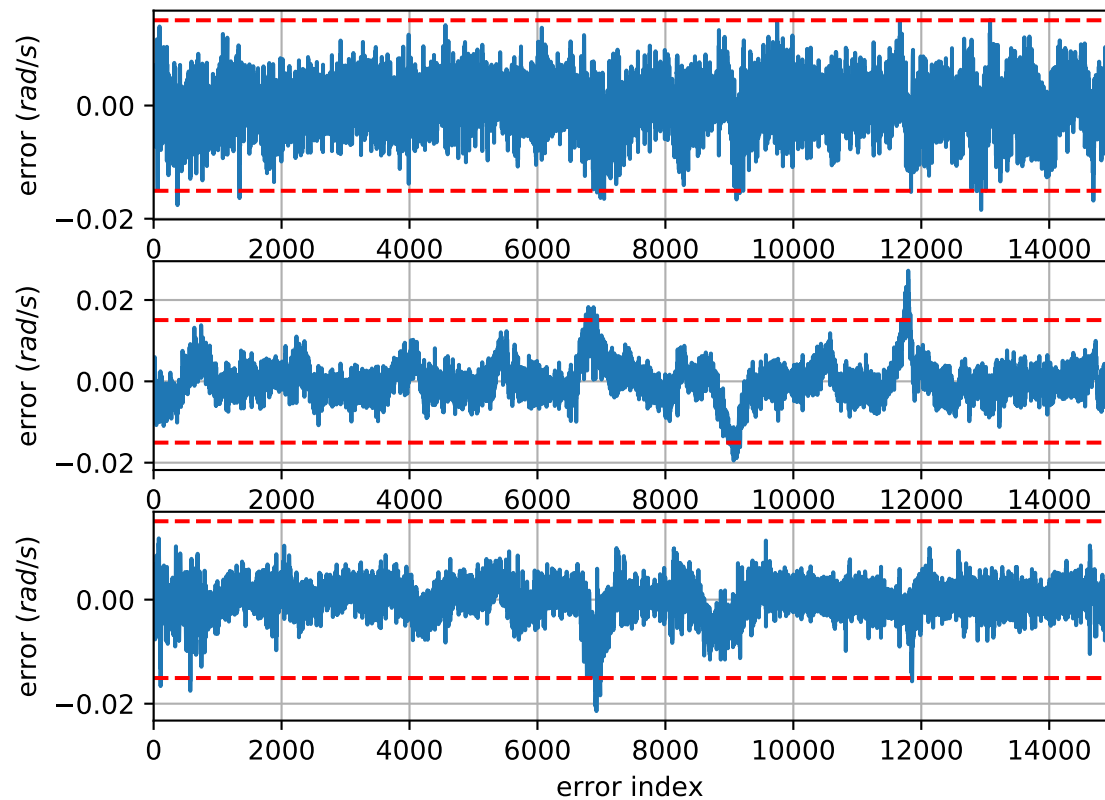
imu0: estimated accelerometer bias (imu frame)



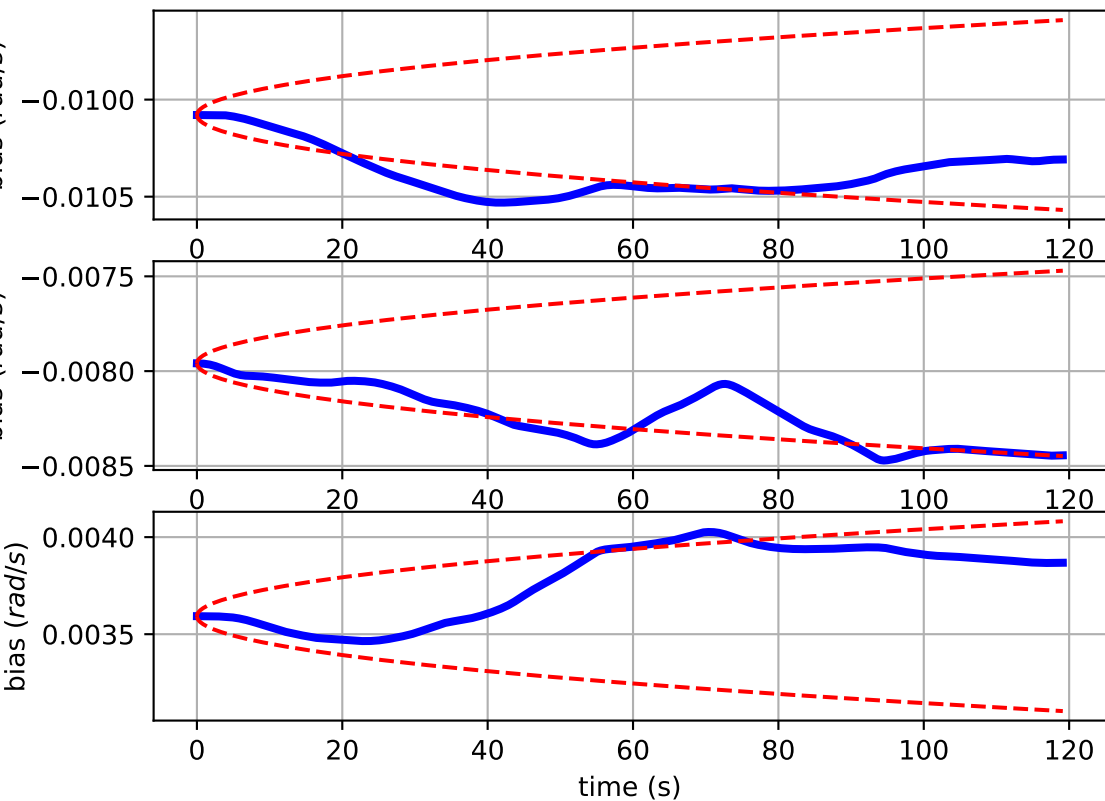
Comparison of predicted and measured angular velocities (body frame)



imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors

